

Generation as Computation

Generalised Computation in the Set-theoretic Universe and Beyond (Thesis Proposal)

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Outline

1 Introduction

- Background: Infinitary Computation
- Background: Formulas as Programs
- Overview

2 Generalised Computation Within V

- Sequential Algorithms
- Generalised Sequential Algorithms
- Relative Computability

3 Generalised Computation Beyond V

- Degrees of Small Extensions
- Local Method Definitions

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Gödel's Informal Proposition

- In an essay Gödel wrote in 1958 (but only published posthumously) [4], he criticised Hilbert's choice of “intuitively clear” notions meant to make up the bedrock of classical mathematics.
- He lamented that “the domain of ‘finitary mathematics’” is unnaturally restrictive.
- In the same work, Gödel informally described a notion of computation built inductively from natural numbers, a notion of which power goes beyond conventional finitary computation.
- He called this notion *computable functions of finite type*.

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More Formal and Explicit Conceptions

- The eponymous Blum-Shub-Smale machines (BSSMs) [5], born from the study of dynamical systems.
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In Classical Computability Theory

- “A program is a formula (of a particular form)” — an oft-repeated slogan.
- An oracle machine can functionally be represented as a sufficiently simple parametrised formula in the language of arithmetic.
- This is a philosophical upshot of relating the arithmetical hierarchy to certain ideals of the Turing degrees.

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Implicit Conceptions of Infinitary Computation

- α -recursion theory, born from ideas by Takeuti [1], Kreisel and Sacks [2] and some others, in the 1960s.
- E -recursion theory, formulated by Normann [3] in the 1970s.
- These are implicit models of computation because they define what it means to be (relatively) computable without explicitly defining what a computation entails.
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In Greater Generality...

- Relative computability is but a form of parametrised definability over some structure.
- Vanilla computability is a special case whereby the parameter in question contains the least amount of information (e.g. using the empty set as parameter).
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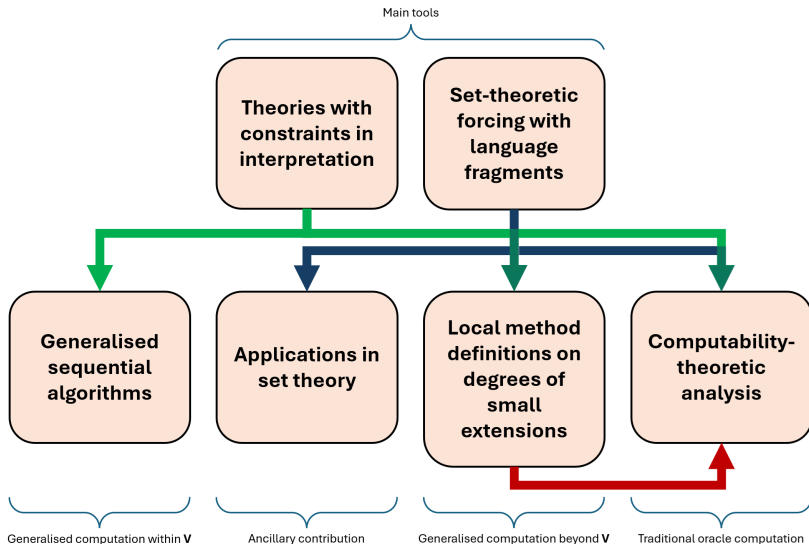
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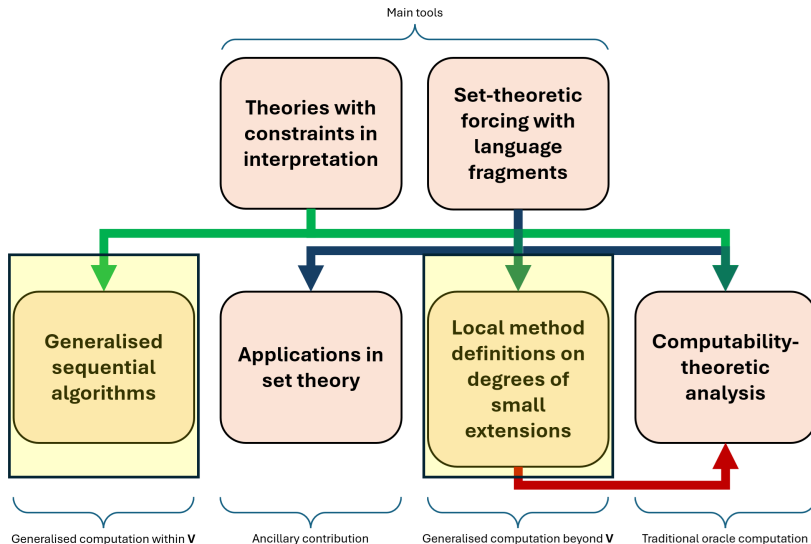
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Overview Chart



Today's Focus



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Higher-Level Descriptions

- The explicit models of computation mentioned so far (BSSMs, ITTMs, etc.) are rather low-level and relatively concrete.
- Computer scientists were interested in more inclusive descriptions of algorithms.
- We want to be able to describe programs in pseudocode, instead of being bound to a fixed programming language.
- But what are the confines of pseudocode?
- To formally answer this question, we need to define things at a higher-level of abstraction.

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- Moschovakis came up with a theory of recursors in an attempt to answer what it means for two programs to be equivalent.
- Gurevich gave axioms (postulates) that a representation of a program should satisfy.
- Gurevich calls any such representation an *algorithm*.
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An Equivalent Definition

Gurevich's formulation involves proper classes, but it can be adapted to avoid size issues.

Definition

- A *sequential algorithm* comprises a set of *states*, including *initial states*, and a *transition function* mapping one state to another.
- Every state in a sequential algorithm is a first-order structure with the same base set and finite signature.
- In identifying the next state, the transition function determines changes to the current state based on finite information about the current state (*bounded exploration*).

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Generality and Intuitiveness

- States are analogous to Turing machine configurations.
- An initial state corresponds to a starting configuration upon receiving an input.
- A final state corresponds to an end configuration from which the output can be read.
- A transition function parallels that of a Turing machine: it codes how the machine behaves.
- Incidentally, this general framework of

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To the Infinite

- For an abstract computation model, sequential algorithms are simple to describe and very inclusive.
- It seems like the go-to candidate for generalisation, if we want an flexible and high-level model of infinitary computation.
- Two principles, isolated by Sieg in [8] as integral to classical models of computation, guide our search for the “correct” infinitary version of sequential algorithms.
 - *Boundedness*: “there is a fixed finite bound on the number of configurations a computer can immediately recognise”.
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Redefining a Transition Function

- We want to talk about local features between two states.
- Since they are first-order structures with the same signature and base set, local features are synonymous with properties given through the truth predicate $\models$.
- We modify the predicate into $\models_2$ so that we can talk about any two such structures simultaneously.
- Every *generalised sequential algorithm* ($GSeqA$) has its transitions governed by a transition formula: transition from s_1 to s_2 iff $(s_1, s_2) \models_2 \phi$ for transition formula ϕ .

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Redefining a Transition Function

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- It suffices but is often not easy to verify.
- By restricting transition formulas, we can formalise bounded exploration locally.
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Ordinal Base Sets

- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
- More precisely, every GSeqA $\mathfrak{M}$ has an associated limit ordinal $\kappa(\mathfrak{M})$.
- Every state of a GSeqA $\mathfrak{M}$ is an expansion of the structure $(\kappa(\mathfrak{M}); \in)$ over a fixed signature $\sigma(\mathfrak{M})$ associated with $\mathfrak{M}$.
- This simplification is inspired by the fact that every set can be coded as a set of ordinals.
- The base set being an ordinal also fits the basic intuition of sequential memory when one thinks about computation.

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- To shift the burden of programming to the definition of the transition formula, we streamline the definitions of other components of a GSeqA.
- We specify the base set of any state to be a limit ordinal.
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- Let $\mathfrak{M}$ be a GSeqAP.
- For each declared variable $\ulcorner X \urcorner \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\phi^{\ulcorner X \urcorner}$ of some specific form over $\sigma(\mathfrak{M})$.
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Initial States

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- For each declared variable $\lceil X \rceil \in \sigma(\mathfrak{M})$, we associate it with a Δ_0 formula $\psi^{\lceil X \rceil}$ of some specific form over $\in$.
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- If $A \subset \kappa(\mathfrak{M})$, then there is a unique initial state s of $\mathfrak{M}$ such that s interprets In as A , Out as $\emptyset$ and $s \models \phi_{\mathfrak{M}}^d$ of $\mathfrak{M}$.
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A GSeqAP $\mathfrak{M}$ is a tuple

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satisfying the properties we have mentioned.

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Outline

1 Introduction

- Background: Infinitary Computation
- Background: Formulas as Programs
- Overview

2 Generalised Computation Within V

- Sequential Algorithms
- Generalised Sequential Algorithms
- Relative Computability

3 Generalised Computation Beyond V

- Degrees of Small Extensions
- Local Method Definitions

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- We want the definition of a limit state to depend locally on the states that come before it in a run.
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- Under this constraint, we show that with enough space (increasing $\kappa(\mathfrak{M})$ if necessary), taking limit/limsup/liminf wherever possible “pointwise” can simulate all other definitions of a limit state.

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A *terminating run* of a GSeqAP $\mathfrak{M}$ is a function from some successor ordinal δ into $S(\mathfrak{M})$ such that

- $Sq(0)$ is an initial state,
- $Sq(\alpha + 1) = \tau_{\mathfrak{M}}(Sq(\alpha)) \neq Sq(\alpha)$ for all $\alpha < \delta - 1$,
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- $\tau_{\mathfrak{M}}(Sq(\delta - 1)) = Sq(\delta - 1)$.

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Relative Computability Relations

Definition (L.)

- A GSeqAP *computes* B from A iff it has a run that starts with In interpreted as A and terminates with Out interpreted as B .
- $B \leq_{\kappa}^P A$ iff some GSeqAP $\mathfrak{M}$ with $\kappa(\mathfrak{M}) = \kappa$ computes B from A .
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If R is one of the above relations, then a set x being R -computable means $x R \emptyset$.

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Let A, B be sets of ordinals. Then $B \leq^P A$ iff $B \in L[A]$.

- This means the degrees of relative computability derived from $\leq^P$ are exactly the degrees of constructibility.
- The very restricted (at first glance) programming we allow in GSeqAPs actually has the full power of transfinite recursion.
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The Relations $\leq_{\kappa}^{P,s}$ and $\leq_{\kappa}^P$

- An ordinal κ is *admissible* iff $(L_{\kappa}; \in)$ is a model of Kripke-Platek set theory.
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Outline

1 Introduction

- Background: Infinitary Computation
- Background: Formulas as Programs
- Overview

2 Generalised Computation Within V

- Sequential Algorithms
- Generalised Sequential Algorithms
- Relative Computability

3 Generalised Computation Beyond V

- Degrees of Small Extensions
- Local Method Definitions

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- What do we mean by “sets outside V ”?
- Step out of V and treat V as a countable transitive model of ZFC (henceforth denoted CTM).
- Look at those sets that can be adjoined to V to give a nice CTM.
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Outer Models

- Let U_1 and U_2 be CTMs. U_2 is an outer model of U_1 iff
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- The binary relation “being an outer model of” is transitive.

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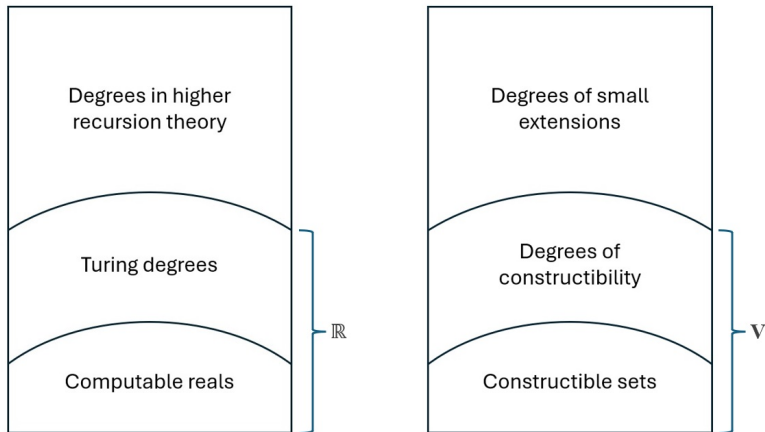


Figure: comparison between conventional notions of relative computability (left) and our generalised notions (right).

Multiverse of Generic Extensions

Definition

Let V be a CTM. The *outward generic multiverse centred at V* is the set

$$\mathbf{M}_F(V) := \{W : W \text{ is a forcing extension of } V\}.$$

The following fact is basic.

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Outline

1 Introduction

- Background: Infinitary Computation
- Background: Formulas as Programs
- Overview

2 Generalised Computation Within V

- Sequential Algorithms
- Generalised Sequential Algorithms
- Relative Computability

3 Generalised Computation Beyond V

- Degrees of Small Extensions
- Local Method Definitions

Referring to Small Extensions in V

- Often it is useful to refer to small extensions of V within V .
- Such references in general cannot isolate any non-trivial small extension.
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- *Theories with constraints in Interpretation (TCIs)*, and their models, are a formalisation of this idea.
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- TCIs provide bounds to the interpretation of a theory T over a signature σ .
- A TCI $\mathcal{T}$ expands on T by specifying additional requirements on a possible model $\mathfrak{M}$ of T .
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Consistency of TCIs

Definition (L.)

A TCI is *consistent* iff it has a model in some outer model of V .

By examining a suitable forcing notion in V , we can decide if a TCI in V is consistent.

Proposition

Let $\mathcal{T}$ be a TCI. Then for all sufficiently large cardinals λ ,

$$\Vdash_{Col(\omega, \lambda)} \exists \mathfrak{M} \text{ (“}\mathfrak{M} \text{ is a model of } \mathcal{T}\text{”) } \iff \mathcal{T} \text{ is consistent.}$$

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Local Method Definitions

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A *local method definition* of V is a non-empty class of TCIs definable in V .

- In the meta-theory, define a function Eval^V from the set of TCIs $\mathcal{T} \in V$ into the set of small extensions of V , such that

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If X and Y are local method definitions of V , $X \leq^M Y$ denotes the statement

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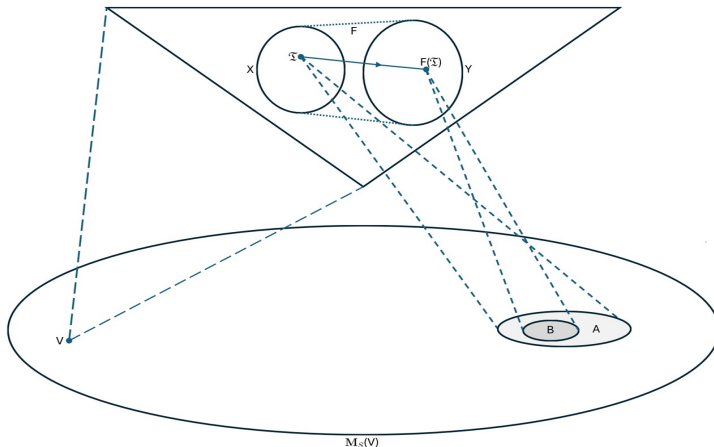


Figure: Visual representation of a function F witnessing $X \leq^M Y$, where X and Y are local method definitions.

The Local Method Hierarchy

- We can also define the complexity of a TCI by just looking at its first-order theory.
- For example, a Σ_n TCI has a first-order theory containing only Σ_n sentences.
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Theorem (L.)

Let $1 \leq n < \omega$. Then for every $\mathcal{T} \in \Pi_{n+1}^M$ there is $\mathcal{T}' \in \Sigma_n^M$ such that

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Corollary

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Set Forcing

- There is an obvious and uniform way of representing any forcing notion $\mathbb{P}$ as a Π_2 TCI $\mathcal{T}(\mathbb{P})$.
- Thus set forcing, as a technique of accessing small extensions of V , can be represented by the local method definition

$$\text{Fg} := \{\mathcal{T}(\mathbb{P}) : \mathbb{P} \text{ is a forcing notion}\}.$$

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By adapting our framework for constructing forcing notions to the context of TCIs, we associate with each Π_2 TCI $\mathcal{T}$ a forcing notion $\mathbb{P}(\mathcal{T})$, such that a unique model of $\mathcal{T}$ can be read off every $\mathbb{P}(\mathcal{T})$ -generic filter over V .

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Let $\emptyset \neq X \subset \mathbf{M}_S(V)$ be defined by

$$X = \{V[x] : \text{"}x \text{ is a } Y\text{-generator"}\},$$

where " X is a Y -generator" has a sufficiently absolute definition.
Moreover, assume there is a set $z \in V$ such that

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- The hypotheses of the previous theorem are a conservative attempt at formalising what a definition of a class of small extensions of V through their generators should entail, if the said definition depends locally on bounded amount of information in V .
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Let $n < \omega$, $\vec{y} \in V^n$, and φ be a $(n+1)$ -ary Δ_1 formula in the Lévy hierarchy. Define

$$U(\vec{y}, \varphi) := \{V[x] : \exists x' \in V (x \subset x') \text{ and } \varphi(\vec{y}, x)\}.$$

If $U(\vec{y}, \varphi) \neq \emptyset$, then there is a forcing notion $\mathbb{P} \in V$ for which $\text{Eval}^V(\mathfrak{T}(\mathbb{P})) \subset U(\vec{y}, \varphi)$.

- Known methods of generating small extensions of a ground model are Δ_1 -definable with the ground model and its members as parameters.
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- The proofs of the previous theorem depends on the maximality principle in an essential manner.
- There is no known way to circumvent that requirement even if we were to restrict the context to showing $\{\mathcal{T}\} \leq^M \text{Fg}$ for $\mathcal{T} \in \Sigma_2^M$.
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Let λ be an infinite cardinal, $1 \leq n < \omega$, and have $\Vdash_{\text{Col}(\omega, \lambda)}^n$ denote the restriction of $\Vdash_{\text{Col}(\omega, \lambda)}$ to Σ_n sentences. If for some $k < \omega$, there is a definable (with parameters) class function F on Σ_k^M such that $\text{ran}(F) \subset V^{\text{Col}(\omega, \lambda)}$ and

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Further Results

- We can refine the function witnessing $\Pi_2^M \leq^M \text{Fg}$ by iteratively pruning the $\mathbb{P}(\mathcal{T})$ s of atoms.
 - This pruning process is analogous to taking Cantor-Bendixson derivatives.
- There are analogues of the previous theorem in V pertaining to certain countable Π_2 TCl's.
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Some Open Questions

- (Q1) What structural properties of $(\mathbf{M}_F(L), \subset)$ also hold for $(\mathbf{M}_S(L), \subset)$?
- (Q2) For example, is $(\mathbf{M}_S(L), \subset)$ downward directed?
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References I

- [1] Gaisi Takeuti. “On the recursive functions of ordinal numbers”. In: *Journal of the Mathematical Society of Japan* 12.2 (1960), pp. 119–128.
- [2] Georg Kreisel and Gerald E Sacks. “Metarecursive sets”. In: *The Journal of Symbolic Logic* 30.3 (1965), pp. 318–338.
- [3] Dag Normann. “Set recursion”. In: *Studies in Logic and the Foundations of Mathematics*. Vol. 94. Elsevier, 1978, pp. 303–320.
- [4] Kurt Gödel. *Collected Works, Volume 2: Publications 1938-1974*. Oxford, England and New York, NY, USA, 1986.

References II

- [5] Lenore Blum, Mike Shub, and Steve Smale. “On a theory of computation and complexity over the real numbers: NP-completeness, recursive functions and universal machines”. In: *Bulletin of the American Mathematical Society* 21.1 (1989), pp. 1–46.
- [6] Joel David Hamkins and Andy Lewis. “Infinite time Turing machines”. In: *The Journal of Symbolic Logic* 65.2 (2000), pp. 567–604.
- [7] Peter Koepke and Martin Koerwien. “Ordinal computations”. In: *Mathematical Structures in Computer Science* 16.5 (2006), pp. 867–884.
- [8] Wilfried Sieg. “Gödel on computability”. In: *Philosophia Mathematica* 14.2 (2006), pp. 189–207.

References III

- [9] Keng Meng Ng, Nazanin R Tavana, and Yue Yang. “A recursion theoretic foundation of computation over real numbers”. In: *Journal of Logic and Computation* 31.7 (July 2021), pp. 1660–1689. ISSN: 0955-792X. DOI: 10.1093/logcom/exab042. eprint: <https://academic.oup.com/logcom/article-pdf/31/7/1660/40820416/exab042.pdf>. URL: <https://doi.org/10.1093/logcom/exab042>.

Thank You!